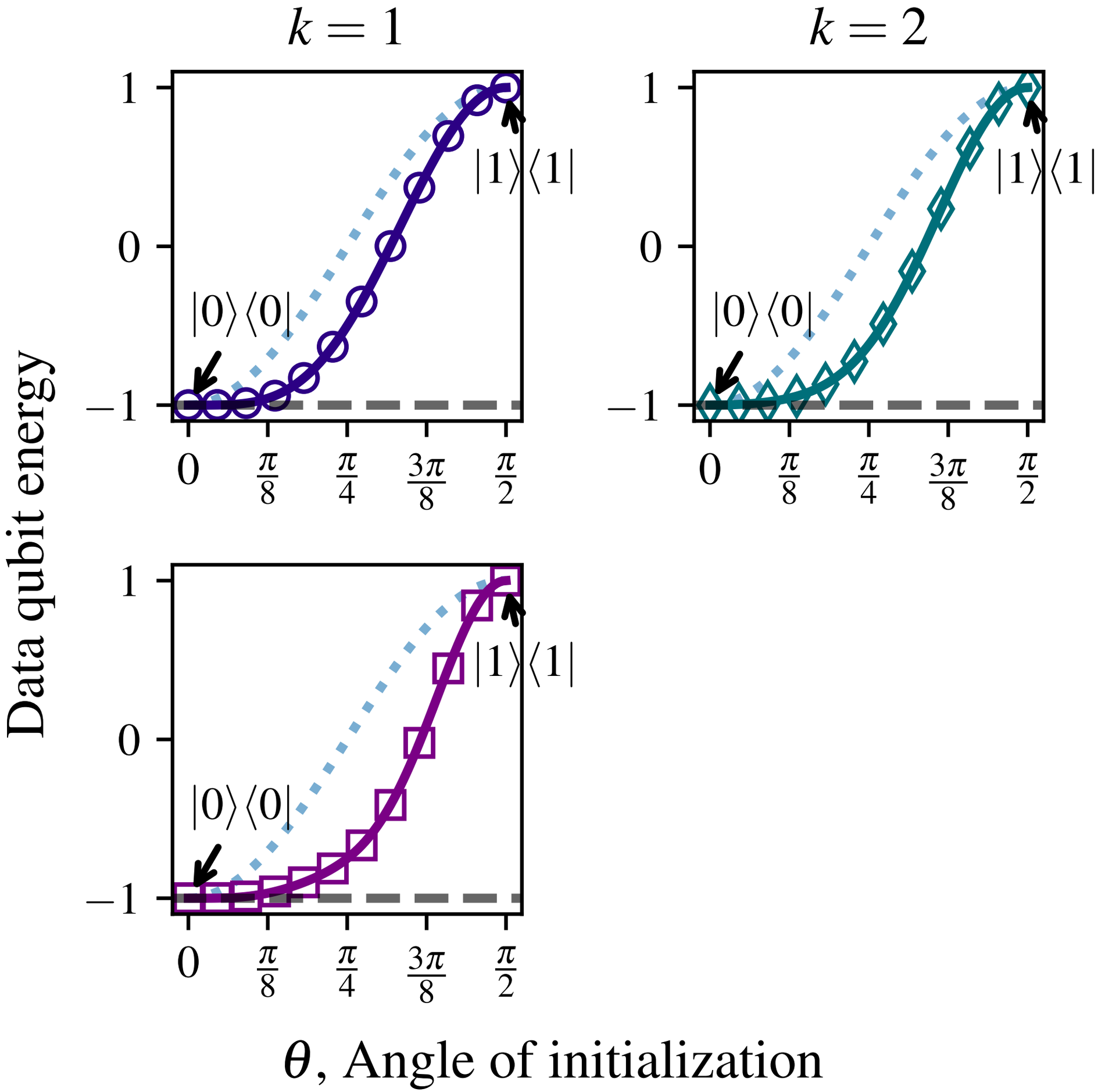

$k = 1$
$k = 2$
1
0
−1
$|1\rangle\langle 1|$
$|0\rangle\langle 0|$
0
$\frac{\pi}{8}$
$\frac{\pi}{4}$
$\frac{3\pi}{8}$
$\frac{\pi}{2}$
Data qubit energy
θ, Angle of initialization